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Administrator: C:\Windows\system32\cmd.exe - locust --host=http://www.ijccr.com
E:\Python27>locust --host=http://www.ijccr.com
[2018-05-10 21:30:48,632] kumargaurav-PC/INFO/locust.main: Starting web monitor
at *:8089
[2018-05-10 21:30:48,634] kumargaurav-PC/INFO/locust.main: Starting Locust 0.8
[2018-05-10 21:31:05,823] kumargaurav-PC/INFO/locust.runners: Hatching and swarm
ing 10 clients at the rate 10 clients/s...
[2018-05-10 21:31:06,838] kumargaurav-PC/INFO/locust.runners: All locusts hatched: websiteUser: 10
[2018-05-10 21:31:06,838] kumargaurav-PC/INFO/locust.runners: Resetting stats
[2018-05-10 21:31:59,920] kumargaurav-PC/INFO/locust.runners: Hatching and swarm
ing 90 clients at the rate 100 clients/s...
[2018-05-10 21:32:01,391] kumargaurav-PC/INFO/locust.runners: All locusts hatched: websiteUser: 90
[2018-05-10 21:32:01,391] kumargaurav-PC/INFO/locust.runners: Resetting stats

```

Figure 3: Executing Locust code to test the Web application

- Mail protocols
 - SMTP
 - POP
 - IMAP
- Web protocols
 - HTTP
 - HTTPS
- Java objects
- Shell scripts
- Native commands

Apache JMeter is a feature-rich IDE for testing, debugging and recording test plans so that the in-depth analytics of the Web applications can be done. The detailed documentation and tutorials are available on the official website of Apache JMeter, which can be used for assorted case studies.

Apache JMeter can be downloaded from http://ApacheJMeter.apache.org/download_ApacheJMeter.cgi for different platforms and flavours as it is cross-platform software. From this link, Apache JMeter is available in compressed format. It is extracted with the executable files which are available in the *bin* directory (Figure 4).

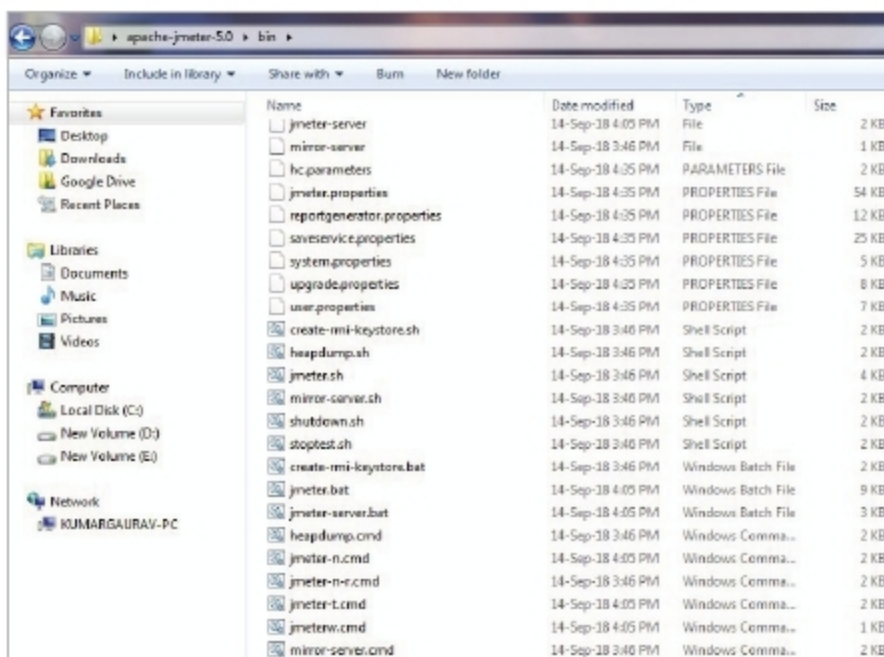
Figure 4: View of executable files in the *bin* folder of Apache JMeter

Table 2: Executable files for Apache JMeter

Operating system	File for executing Apache JMeter
Windows	<i>Apache JMeter.bat</i>
Mac	<i>Apache JMeter.sh</i>
Linux	<i>Apache JMeter.sh</i>

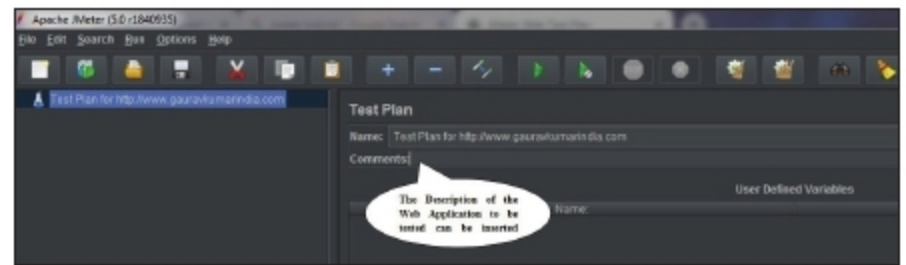


Figure 5: Labelling or naming a test plan in Apache JMeter

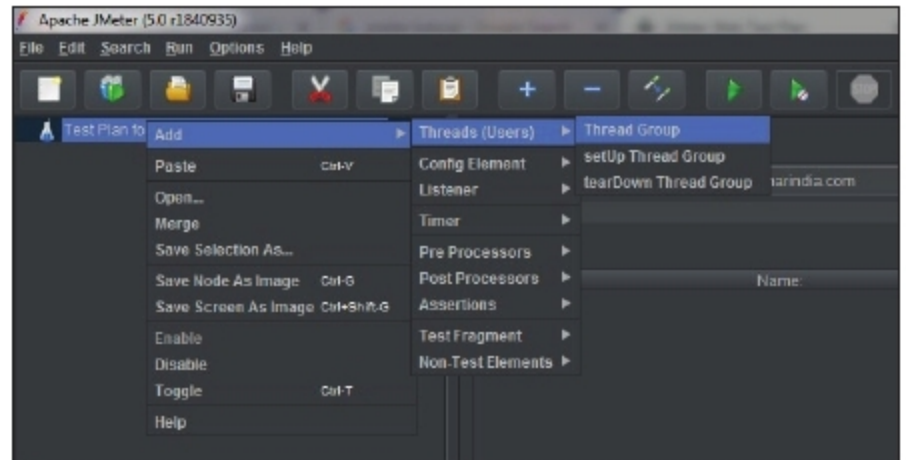


Figure 6: Creating a thread group for load testing

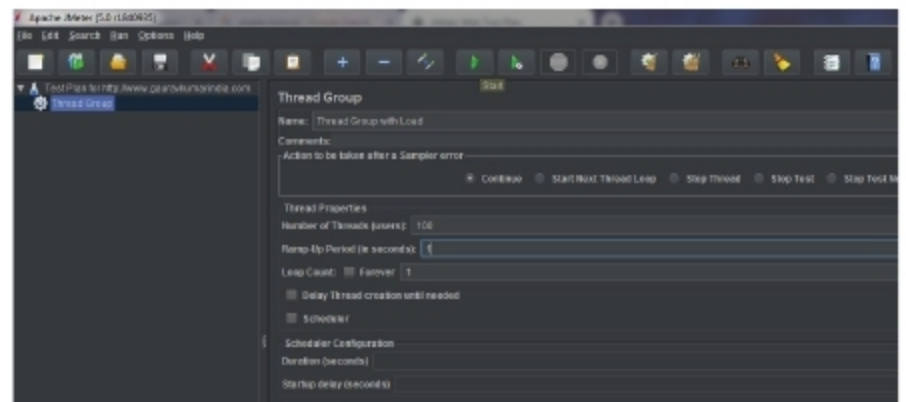


Figure 7: Specifying threads (users) in Apache JMeter

The GUI of Apache JMeter has many features and options for creating and evaluating test plans.

There is no specific naming convention or compulsion regarding the title of a test plan. The testing engineers or audit experts can specify the name or title as per their own preferences (Figure 5).

The thread group is used to generate the voluminous traffic that is required for the testing of an application (Figure 6).

A number of users are created virtually without a physical presence so that the Web application can be tested and, thereby, the cumulative performance can be investigated by the Web masters and administrators.

The option of *HTTP Request* is selected because in the case of Web application testing, the HTTP protocol works to present the outcome on the Web browser or HTTP client. The HTTP client opens the Web page or URL only if that particular link is responding without any issues.

The Web application under testing can be specified in the suite of Apache JMeter in terms of the official link or IP address so that the direct connectivity with the Web server can be evaluated for further investigations (Figure 9).

The option to select 'View Results Tree' enables the Web administrator to visualise whether the load testing parameters